Keysight PROPSIM

Runtime Emulation

User Guide

KEYSIGHT TECHNOLOGIES

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Manual Part Number

F8800-93110

Revision

Revision 2.1, 29th June 2021

Published by:

Keysight Technologies Finland Oy
Elektroniikkatie 10
90590 Oulu, Finland

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1 INTRODUCTION

The document explains the functionality and features of the Runtime emulation feature for PROPSIM. This document is written for PROPSIM.

1.1 Option Description

Runtime Emulation option allows the user to control the emulation during the test run. The commands are given as ATE command from external computer controlling the emulation scenario.

Unlike in a normal file based emulations the Runtime channel models are not pre-calculated but the user can adjust propagation conditions during the test run – on the fly. This is useful in tests where the change in radio propagation conditions effect how the scenario evolves in future. The Runtime control feature is useful also in automated test environments, or generally in any place where the previous test result has an effect to next test condition.

1.2 Key Features

- Allows to control channel model parameters during the emulation run
- Enables to run scenarios where the current radio channel state has impact on how the scenario progresses in future
- Allows to control the emulation by external scenario based software
- Enables extremely long evolving emulations

1.3 Specifications

- Controllable parameters for each channel separately
  - Amplitude, Delay, Doppler, Environment
- Amplitude control range: 60 dB + signal off
- Delay control range: 2.4 us up to 160 us
- Doppler control range 0...200 kHz (not available in F64 release 1.0)
- Number of paths per channel up to 24
- Number of logical channels up to 1024 per emulator
- Runtime command rate: More than one per second.
- Supported fading profiles:
  - Constant
  - Classical (Rayleigh fading, Jakes Doppler)
  - Pure Doppler
  - Rician
  - Flat Doppler
- Supported interfaces: LAN

1.4 Document history

The following table lists the main changes to issues of this document.

Issue	Date	Summary of Changes
1.0		Release of PROPSIM Runtime Emulation

Issue	Date	Summary of Changes
1.1	Dec 2014	New documentation template.
1.2	Jan 2016	Template updated.
1.3	Jan 2018	New documentation template.
1.4	Jun 2019	PROPSIM F64 changes.
2.0	Feb 2020	PROPSIM GUI changes.
2.1	Jun 2021	Customer support information updated.

2 RUNTIME EMULATION MODE

Runtime emulation is an additional operating mode in the PROPSIM. Emulator goes automatically into Runtime operating mode when a Runtime emulation file is loaded.

Runtime emulations are used in ATE (Automatic Test Equipment) interface only. ATE services of PROPSIM provide the user a method to control the emulator remotely via LAN interface.

PROPSIM ATE interface and commands are described in detail in PROPSIM user reference. This document provides incremental information for Runtime emulation mode.

2.1 Runtime models and controllable parameters

Runtime model file (.RTC) includes a set of user predefined environments. Each environment is defined by given power delay profile (PDP). During the runtime emulation delay and amplitude can be changed and the environments can be swapped between those defined in the RTC file.

The basic idea on how the delay and amplitude controls work is shown in Figure 1. Environment can be selected separately to each channel in the emulation. Controllable parameters are summarized in Table 1.

Table 1. Runtime controllable parameters

Parameter	Description
Amplitude	Link gain, range down to -200 dB and OFF, resolution 0.1 dB. Specified as total channel gain between the emulator input and output ports with minimum attenuation range of 60 dB. Link OFF sets a max attenuation that guarantees a full loss of signal at the intended radio link.
Delay	Range 2.4 us ... 160.0 us, hardware delay resolution is 5, 10 or 20 ns in PROPSIM F64, depending of the emulation bandwidth. This is absolute delay between radios. Minimum delay value depends on hardware insertion delay and channel topology (SISO, MIMO, MANET/Mesh, ...).
Doppler	The velocity of a link is given as Doppler. This is Doppler as seen by the intended receiver from certain transmitter. It is a relative value since both TX and RX may be in motion with different speed and directions.
Environment	Propagation environment type, defined with power-delay-profiles (PDPs) in RTC file.

Detailed syntax for Runtime commands is described in chapter 4.

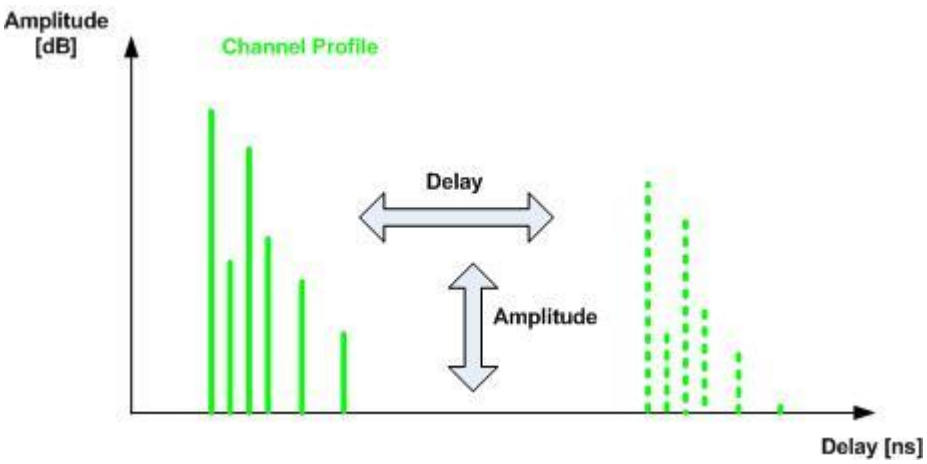

Figure 1. Runtime controlling basics

Runtime model with the filename extension .RTC is a container file which includes several environments (channel profiles). The high level modeling diagram is shown in Figure 2.

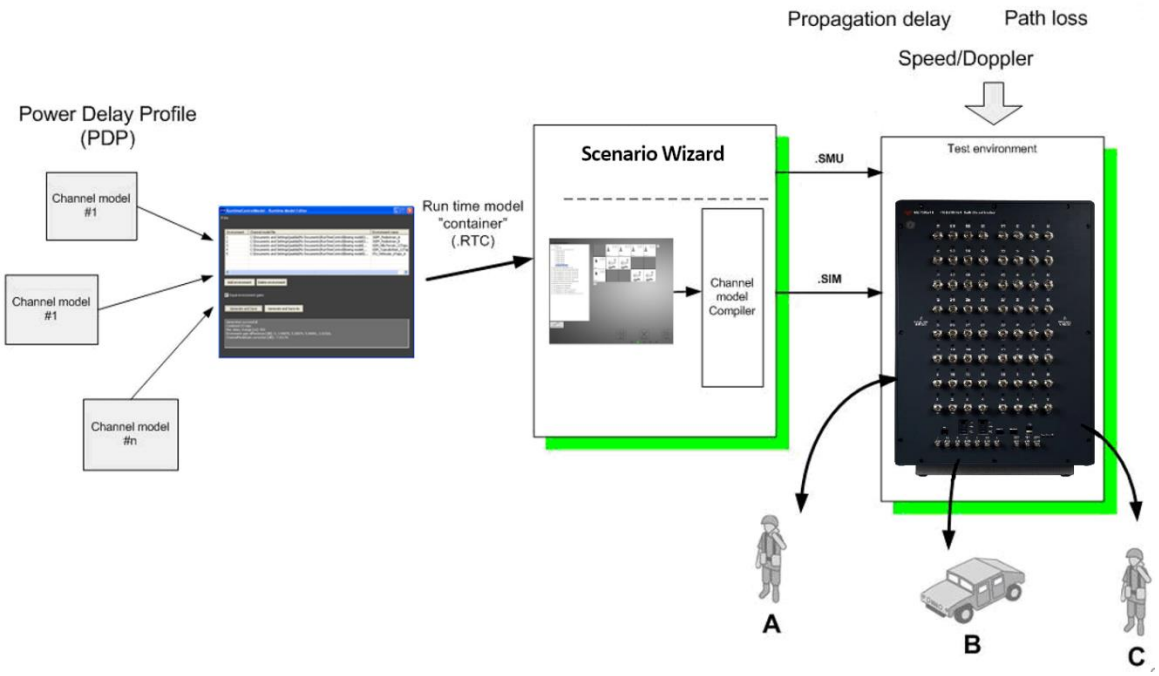

Figure 2: Runtime modelling and emulation generating flow

3 CREATING RUNTIME MODELS

3.1 Runtime Model Editor

RTC editor is an application that is used to merge the channel models (.tap-files) to one runtime channel model (.rtc-file). To open the RTC editor, select **Utilities** > **Runtime Model Editor** in the navigation bar. This is shown in Figure 3.

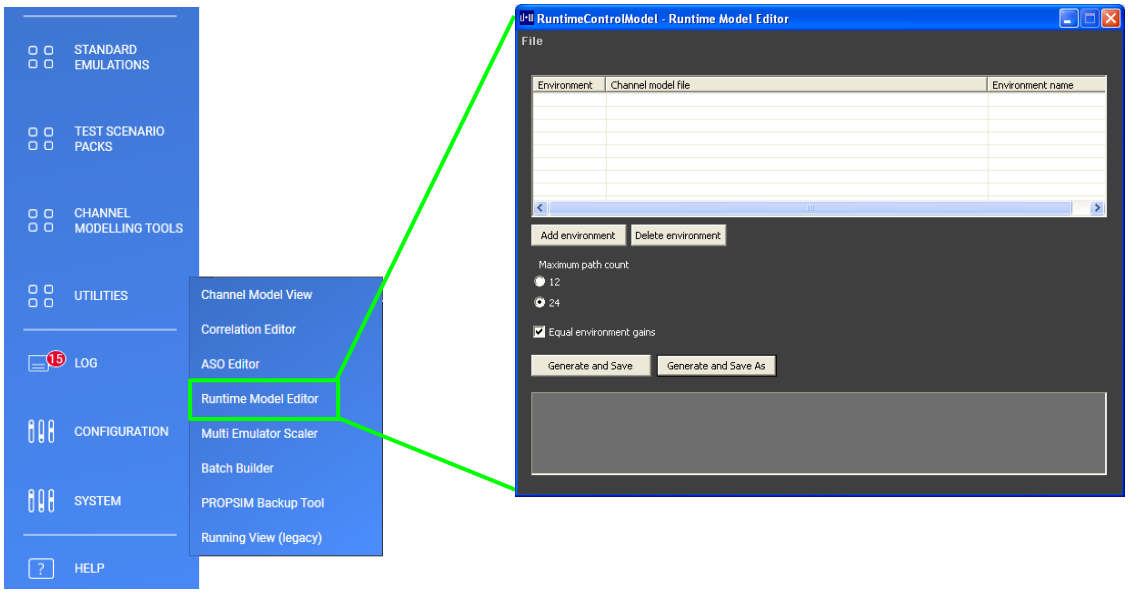

Figure 3. Runtime Model Editor in Main Toolbar

3.1.1 Functions

The first step in creating Runtime model is to choose the models. Select suitable .tap files from PROPSIM channel model library or generate the models by using PROPSIM channel model editor.

File menu contains the following items:

Open existing runtime (.rtc) files

Generate and Save existing rtc file

Generate and Save as saves the existing rtc file with different file name

Exit from the editor application

Add environment button opens a dialog where channel profile .tap files can be selected. When a model is selected it will appear in the model list (Figure 4)

Environment	Channel model file	Environment name
1	C:\models\Runtime\3GPP_Pedestrian_A.tap	3GPP model renamed here
2	C:\models\Runtime\3GPP_Pedestrian_B.tap	3GPP_Pedestrian_B
3	C:\models\Runtime\GSM_HillyTerrain_12Taps_1.tap	GSM_HillyTerrain_12Taps_1
4	C:\models\Runtime\GSM_TypicalUrban_12Taps_1.tap	GSM_TypicalUrban_12Taps_1
5	C:\models\Runtime\ITU_Vehicular_6Taps_A.tap	ITU_Vehicular_6Taps_A

Figure 4. Runtime channel model list (list of environments)

Environment names can be edited directly in the list. When running the models the channel profile can be swapped by using environment number (the first column) or the Environment name. Default name for the model is the file name of tap-model

Delete environment button removes the selected model from the list. If none is selected the last line will be removed.

Runtime environment gains will be normalized when the **Equal environment gains** check box is activated. This means that there won't be jump in power level when the environment is changed during the emulation. If the check box is not selected the gain values written in the channel model (.tap) files will be used as such.

Generate and Save or **Generate and Save As** buttons will activate the model conversion. While the RTC file is generated the log window shows information about the generated model like shown in the example below.

Generation successful

Combined 23 taps

Max delay change [ns]: 500

Enviroment gain differences [dB]: 0, 3.40878, 5.28074, 5.84891, 2.63326,

ChannelModelGain correction [dB]: -7.8117

The maximum number of paths per channel in runtime models is 12 or 24, depending on the available channel count in emulation and bandwidth. Channel count and bandwidth values are selected in Editing view when emulation is created. Select **Maximum path count** so that it is equal to or smaller than the Available paths in Editing view.

In case the total amount of paths in different environments exceeds selected path count value, the RTC application will optimize the accuracy of the models in a way that the emulation can be run in one emulator channel. First the RTC editor application will combine paths which are on same delay. If further simplifications must be done in order to fit the models in one RTC file, the application will round path delays in such a way that the error is minimized. In the example above the maximum delay rounding has been 500 ns.

If the **Equal environment gains** check box is selected the RTC editor will calculate model gains and balance gain values. In the example above max correction made to one channel model is 7.8 dB. Please note that the relative path gains are not changed but just the total gain of the model.

4 CREATING RUNTIME EMULATIONS

The following example shows how to create a runtime emulation.

4.1 Create Runtime Channel Model

Create a runtime channel model using Runtime Model Editor. In this example, file named RuntimeModel.rtc has been created. This model has five environments named similarly as in Figure 4.

4.2 Create Emulation

Open the Scenario Wizard and start creating a new emulation.

- Cellular systems emulations: To add the runtime channel mode to a link, select the link, and in the Link properties window, browse and select the runtime channel model for uplink and downlink.

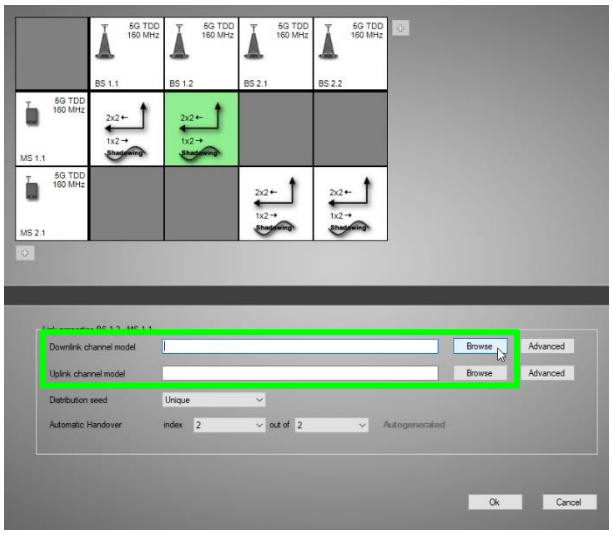

Figure 5 Adding channel model to link in cellular systems emulation

- MANET/Mesh emulations: To add the runtime channel model to all the channels, in step 2/4, browse and select the runtime channel model in the Channel model field. Alternatively, to assign channel-specific channel models, select the Advanced button.

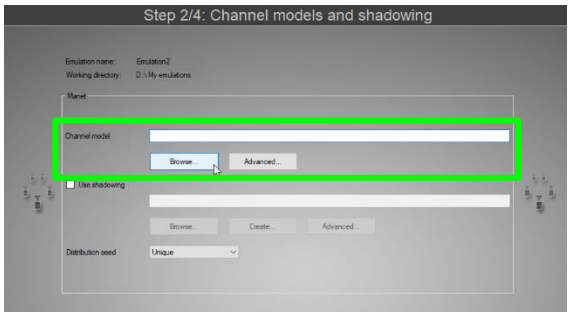

Figure 6 Adding channel model in MANET emulation

If needed, see PROPSIM User Reference for more information.

5 EMULATION CONTROL

This chapter describes how the emulation is loaded and controlled. Information here is incremental to standard PROPSIM ATE operation.

5.1 Emulation loading and running

The following example shows the how to use the runtime control with ATE commands. The example assumes that the emulation is already made as in chapter 4. All of the channels have the same channel model with five different environments. More information and examples about how to initialize the ATE communication interface and how to open and run models can be found from [1].

```
// reset the PROPSIM Emulator
*RST

// load the emulation
CALCulate:FILTer:FILE D:\Emulations\Runtime\3RadioManet_example.smu

// check the physical connector inputs and outputs settings for each
// channel
ROUTe:PATH:CONNector? 1
ROUTe:PATH:CONNector? 2
ROUTe:PATH:CONNector? 3
ROUTe:PATH:CONNector? 4
ROUTe:PATH:CONNector? 5
ROUTe:PATH:CONNector? 6
// based on the information received from the emulator connect the
// physical inputs and outputs accordingly. Alternatively use Active
// connector view (see Figure 7)

// start emulation, initially channels have the first environment loaded
DIAGnostic:SIMUlation:GO

// start test process by sending runtime control commands to one or more
// change a single channel
CH:MOD:CONT:ENV 1,GSM_HillyTerrain_12Taps_1,-38.7,1510006,26169

// instead of the environment name the environment number can be used
CH:MOD:CONT:ENV 1,4,-38.7,1510006,26169

// change multiple channels with a single command
CH:MOD:CONT:ENV 1,4,-31.5,1954366,6138,2,2,-37.3,1740025,17924,3,1,-35.7,3355673,25957,5,2,-
26.8,9244761,25582

// gain can be set to 'off'
CH:MOD:CONT:ENV 1,ITU_Vehicular_6Taps_A,off,6965173,11713

// parameters not wished to be changed can be left empty
CH:MOD:CONT:ENV 1,GSM_TypicalUrban_12Taps_1,,,,4,3GPP_pedestrian,,,

// Ask current status using CH:MOD:CONT:ENV? query
CH:MOD:CONT:ENV? 1
// Response is 1,4,-31.5,1954366,6138


// after the last runtime control command close the emulation
DIAGnostic:SIMUlation:CLOSE
```

5.2 Verifying connections

Runtime emulations are used via ATE interface. However, it might be useful to check connections in local mode. This can be done by loading the model in the Emulation Control View like standard file based models. Now the first environment is loaded and the connections can be checked from the Active connectors view in the Data views window, as shown in Figure 7. To open the Active connector view, click the **Data views** button in the top right corner of the PROPSIM GUI and drag the **Active connectors** view from the data view selection bar into one of the slots in the data view dashboard.

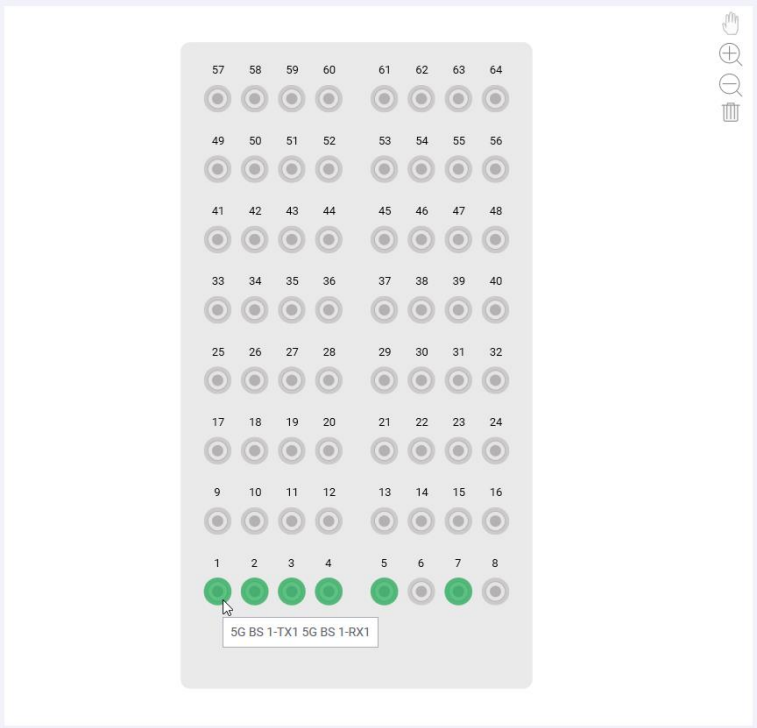

Figure 7. Active connector view showing RF-cable connections (PROPSIM F64)

5.3 Running Runtime emulation

When PROPSIM emulator boots up, both the ATE GPIB Server and the ATE LAN Server are started. As ATE commands are received to either interface, they are displayed in the corresponding monitor console, which also shows possible responses and error messages. The commands how to load and start running emulations are described in Remote Control chapter of PROPSIM User Reference.

5.4 Runtime commands

5.4.1 Environment control command

Syntax:

CHannel:MODel:CONTrol:ENVironment <ChannelNumber>,<Environment>,<gain>,

<delay>,<Doppler>,<ChannelNumber>,<Environment>,<gain>,<delay>,<Doppler>,

... etc, as many channels as needed to be updated.

Description:

This command is used for controlling the radio channel parameters in runtime emulation.

<Channel number>	1 = ch1, 2 = ch2, etc.
<Environment>	Name or number of the environment specified in .rtc file
<Gain>	Channel gain (always with negative sign). given in dB's or off

	This command affects to the whole environment profile, the power ratio between model paths is not affected
<Delay>	Delay of the model, given in ns's
<Doppler>	Doppler between radios (relative speed) , given in Hz's Note: Changing doppler in runtime emulation is not supported in PROPSIM F64 Release 1.0

Examples:

Example 1: updates 3GPP_Pedestrian_A environment parameters to channels 1 and 3:

ch:mod:cont:env 1,3GPP_Pedestrian_A,-20,2000,,3, 3GPP_Pedestrian_A,-40,3000,

Example 2: updates only Delay to channels 2 and 4:

ch:mod:cont:env 2,,,6000,,4,,,7000,

5.4.2 Environment status query

Syntax:

CHannel:MODel:CONTrol:ENVironment? <Channel number>,<Channel number>

Description:

This command gets the status of runtime control parameters in channel.

<Channel number>	1 = ch1, 2 = ch2, etc.

Example:

Get environment status in channels 1 and 3.

ch:mod:cont:env? 1,3

And the result should be

1, 3GPP_Pedestrian_A,-21,2100,123,3, 3GPP_Pedestrian_A,-22,2200,321

6 REFERENCES

[1] PROPSIM User Reference